

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

- This project predicts whether SpaceX Falcon 9 rockets will land successfully. Successful landings reduce costs because rockets can be reused.
- **Methods Used:**
 - Data Collection: Web scraping + APIs
 - Data Cleaning: Removing errors and missing data
 - EDA & Visualization: SQL, Matplotlib, Seaborn, Plotly
 - Geospatial Analysis: Maps with Folium
 - Machine Learning: Logistic Regression, SVM, Decision Tree, Random Forest
- **Results:**
 - Best model accuracy: **83%**
 - Payload and booster type strongly impact success
 - Launch sites visualized on interactive map
 - Final interactive dashboard built with Dash

Introduction:

- **SpaceX** is revolutionizing space travel by reusing the first stage of their Falcon 9 rockets. A successful landing saves millions of dollars per launch. However, predicting whether a rocket will land successfully is a challenging task.
- In this project, we collected and analyzed SpaceX launch data to build a machine learning model that predicts landing success.
- **Problem Statement**
 - Can we predict if a Falcon 9 first stage will land successfully?
 - How do factors like launch site, payload mass, and booster version affect success?
 - Can this model help estimate launch costs and assist competitors in making decisions?

Section 1

Methodology

Methodology

- Data collection methodology:
 - Web scraped launch data from SpaceX official site.
 - Retrieved additional data from **SpaceX APIs**.
 - Stored structured data in pandas DataFrames.
- Perform data wrangling
 - Cleaned missing values and incorrect data types.
 - Formatted columns for analysis (dates, payloads, etc.).
 - Engineered new features for modeling (e.g., binary landing outcome).
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
- Used **Grid Search** to find best model parameters.
- Evaluated models using accuracy scores, precision, recall.
- Selected the **best performing model** for final predictions.

Data Collection

1. Web Scraping:

- Scraped SpaceX launch data using **BeautifulSoup & Requests**.

2. SpaceX API:

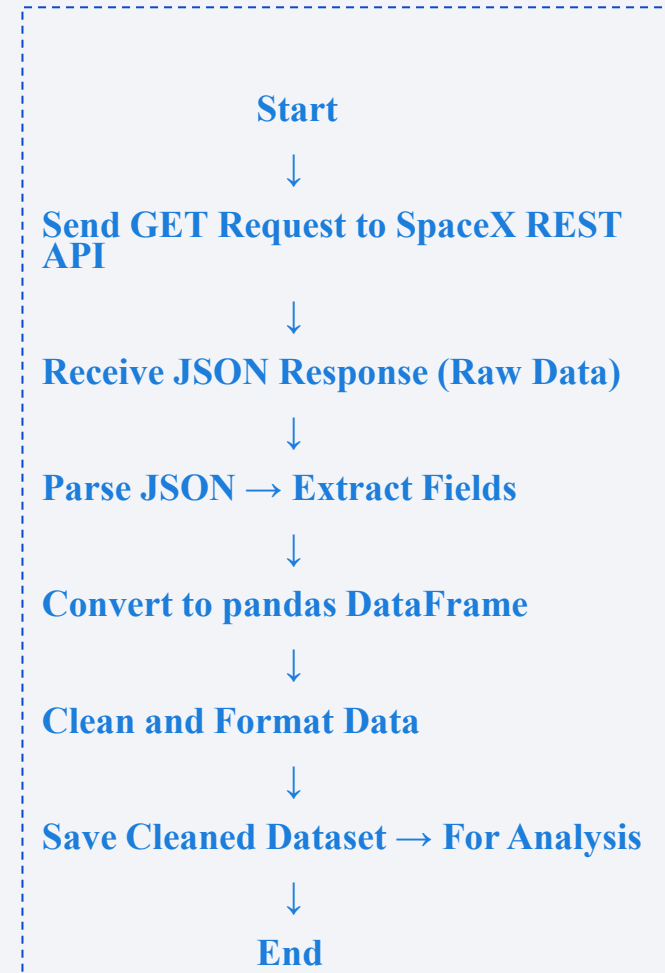
- Fetched detailed launch records like rocket name, payload, and landing outcome via API.

3. Wikipedia Data:

- Extracted additional payload details using **pandas** `read_html()` for missing values.

Data Collection – SpaceX API

- **Sources of Data:**
- **Web Scraping**
Scraped SpaceX launch data from wikipedia using BeautifulSoup and requests library.
- **SpaceX API**
Retrieved detailed launch records using SpaceX API
- **Notebook Link:**
- https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/jupyter-labs-spacex-data-collection.ipynb



Data Collection - Scraping

- **Objective:**
- To collect additional SpaceX launch data by scraping from wikipedia
- **Notebook link:**
- https://github.com/jatharvaO4/SpaceX_DataScience_Project/blob/main/Jupyter-labs-webscraping.ipynb



Data Wrangling

- **Data Wrangling :**
- **Merged** API and web scraping data.
- **Removed** missing values and duplicates.
- **Converted** dates & numbers to correct formats.
- **Created** binary success column for landing outcome.
- **Engineered** features → Booster version, landing type.
- **Final Output:** Clean dataset → used for EDA & machine learning.
- **Notebook link:**
- https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- **EDA with Data Visualization**
- **Scatter Plot:** Flight Number vs. Launch Site → To observe how launches vary by site
- **Scatter Plot:** Payload Mass vs. Launch Site → To analyze payload distribution across launch sites
- **Bar Chart:** Success Rate by Orbit → To compare success rates between orbit types
- **Scatter Plot:** Flight Number vs. Orbit Type → To study launch patterns for different orbits
- **Scatter Plot:** Payload Mass vs. Orbit Type → To see how payload mass relates to orbits
- **Line Chart:** Yearly Average Success Trend → To visualize how SpaceX success rate improved over time
- **Notebook link:**
https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/eda-data-vizualisation.ipynb

EDA with SQL

- **Queried all unique launch site names** → To list launch locations
- **Filtered launch sites starting with 'CCA'**
- **Calculated total payload mass for NASA** → To find NASA's total payload contribution
- **Computed average payload for booster version F9 v1.1** → For model analysis
- **Identified first successful ground landing** → To track SpaceX's landing milestone
- **Listed boosters with successful drone ship landings (Payload 4000–6000 kg)** → For payload analysis
- **Counted successful vs failed missions** → Overall mission success rate
- **Found booster with maximum payload mass** → To discover largest payload missions
- **Queried failed drone ship landings in 2015** → Failure analysis for that year
- **Ranked landing outcomes between 2010 and 2017** → To see most common outcomes

Notebook Link:

https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- **Map Objects Used:**

- Markers (folium.Marker):
→ To show launch site locations on the map.
- MarkerCluster (folium.plugins.MarkerCluster):
→ To group markers together and avoid clutter when zoomed out.
- Color-coded Markers:
→ White markers with colored icons → Represent success (green) or failure (red) of landings.
- Circles (folium.Circle):
→ To show proximity to nearby infrastructure (e.g., roads, coastlines) for specific launch sites.
- Lines (folium.PolyLine):
→ Used to visualize distance measurements from launch sites to nearby features.

- **Why These Objects?**

- To visualize where launches occurred globally.
- To highlight successful vs failed launches visually.
- To show how close launch sites are to key features like railways, highways, or coastlines.
- **Notebook link:** https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- **Interactive Dashboard (Plotly Dash)**
- **Dropdown:** Select launch site
- **Pie Chart:** Shows success ratio
- **Slider:** Filter by payload mass
- **Scatter Plot:** Payload vs. Success (**color** → booster type)
- **Why?** To **explore relationships** between site, payload, and success interactively.
- **Dashboard Notebook:**
- https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/SpaceX-Dash-app

Predictive Analysis (Classification)

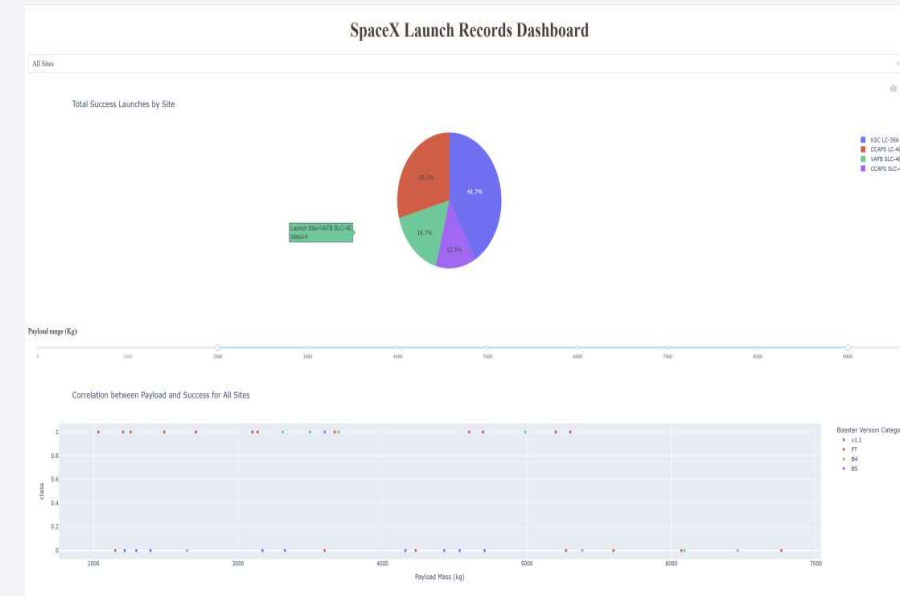
- **Process:**
- **Built multiple models:** Logistic Regression, SVM, Decision Tree, Random Forest
- **Evaluated models** → Used **accuracy score** to compare
- **Improved performance** → Applied **GridSearchCV** for hyperparameter tuning
- **Selected best model** → Based on **highest accuracy**

- **Flow:**
Prepare Dataset → Train Multiple Models → Evaluate Accuracy → Hyperparameter Tuning →
Select Best Model

- **Notebook Link:**
- https://github.com/jatharva04/SpaceX_DataScience_Project/blob/main/SpaceX_Machine%20Learning%20Prediction.ipynb

Results

- **Exploratory Data Analysis Results (Summary)**
 - **Launch site CCAFS LC-40 → Most frequently used launch site.**
 - **Payload range 4000–6000 kg → Highest probability of success.**
 - **F9 FT boosters → Highest success rate overall.**
 - **Trend: Clear improvement in launch success over time**
-
- **Predictive Analysis Results (Summary)**
 - **Best Performing Model: Random Forest Classifier.**
 - **Final Accuracy: ~83%.**
 - **Features Used: Launch site, Payload Mass, Booster Version.**
 - **Conclusion: Model effectively predicts first stage landing success, supporting cost estimation and operational planning.**



Results dashboard demo

SpaceX Launch Records Dashboard

All Sites

100%

100%

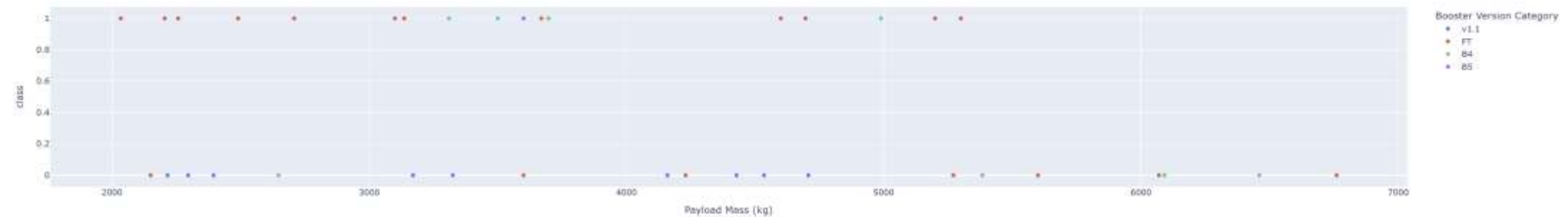
Total Success Launches by Site

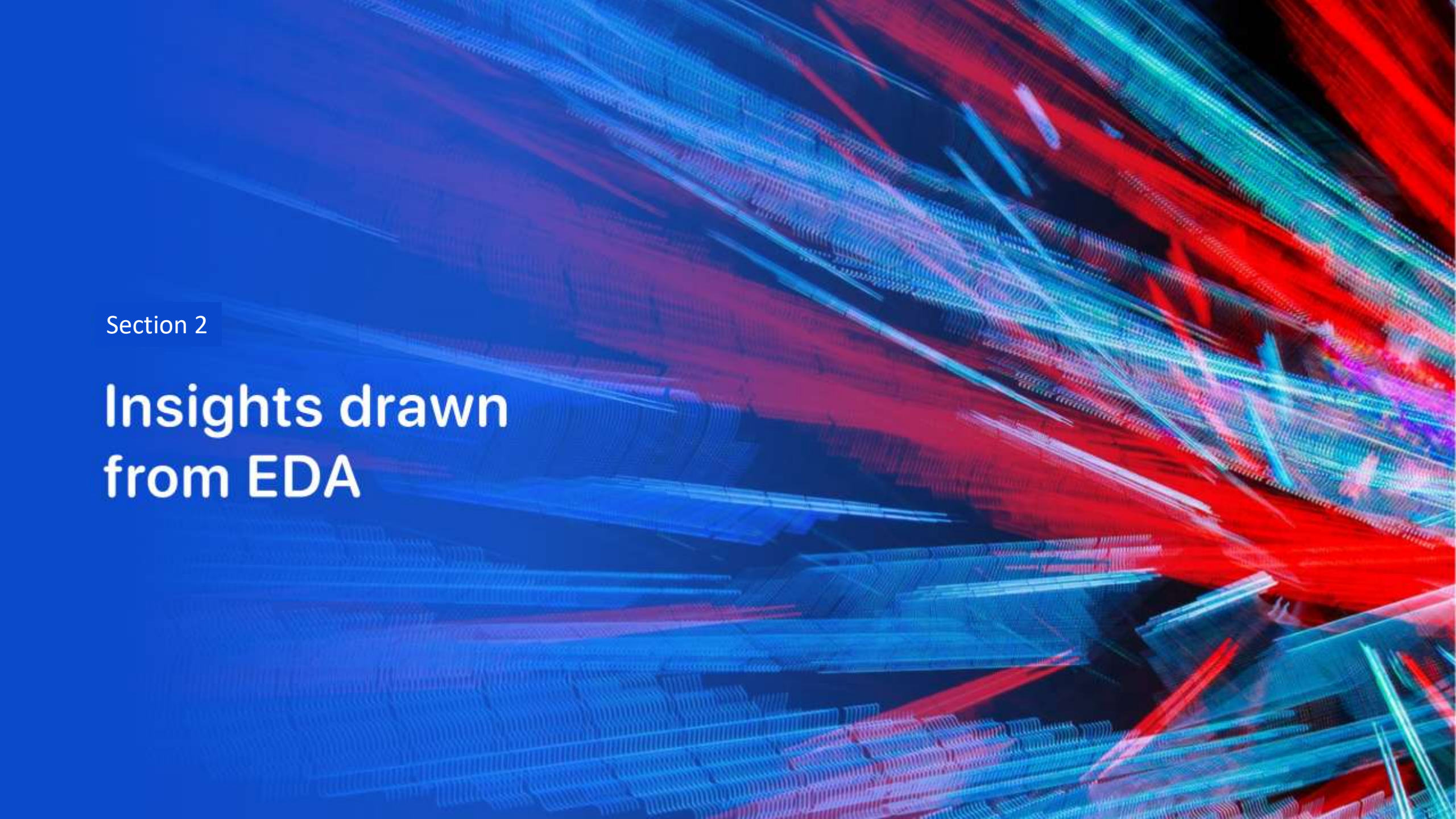


Payload range (Kg):



Correlation between Payload and Success for All Sites



The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue, red, and cyan on the right. Overlaid on these streaks is a faint, semi-transparent grid of small squares, creating a complex, layered visual effect.

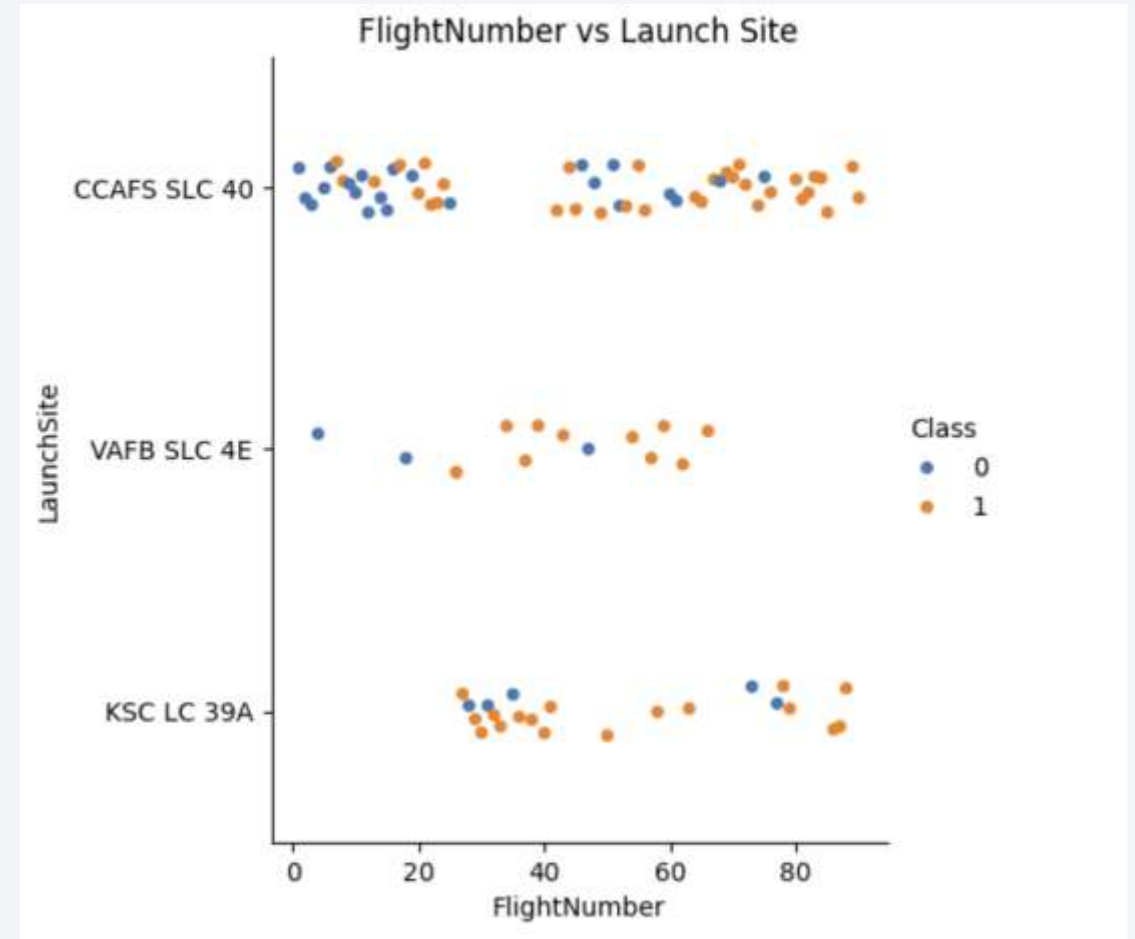
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Explanation:

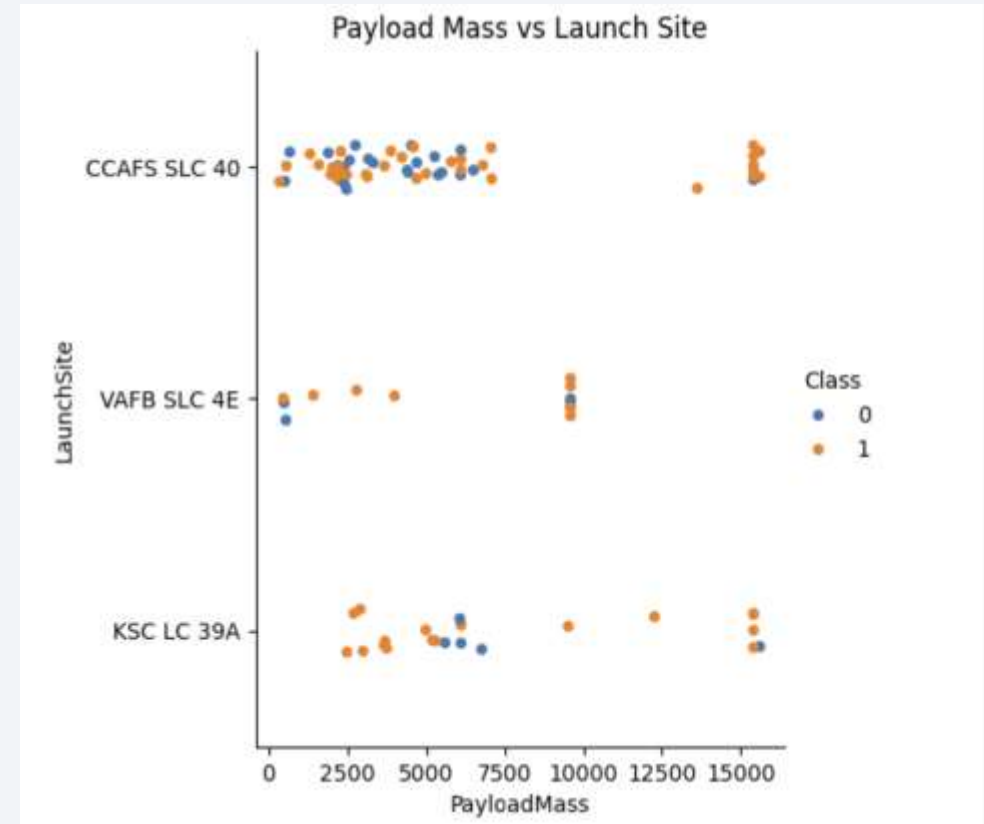
Shows how the number of launches increases over time for each launch site. Helps visualize which sites have been used [more frequently](#).



Payload vs. Launch Site

Explanation:

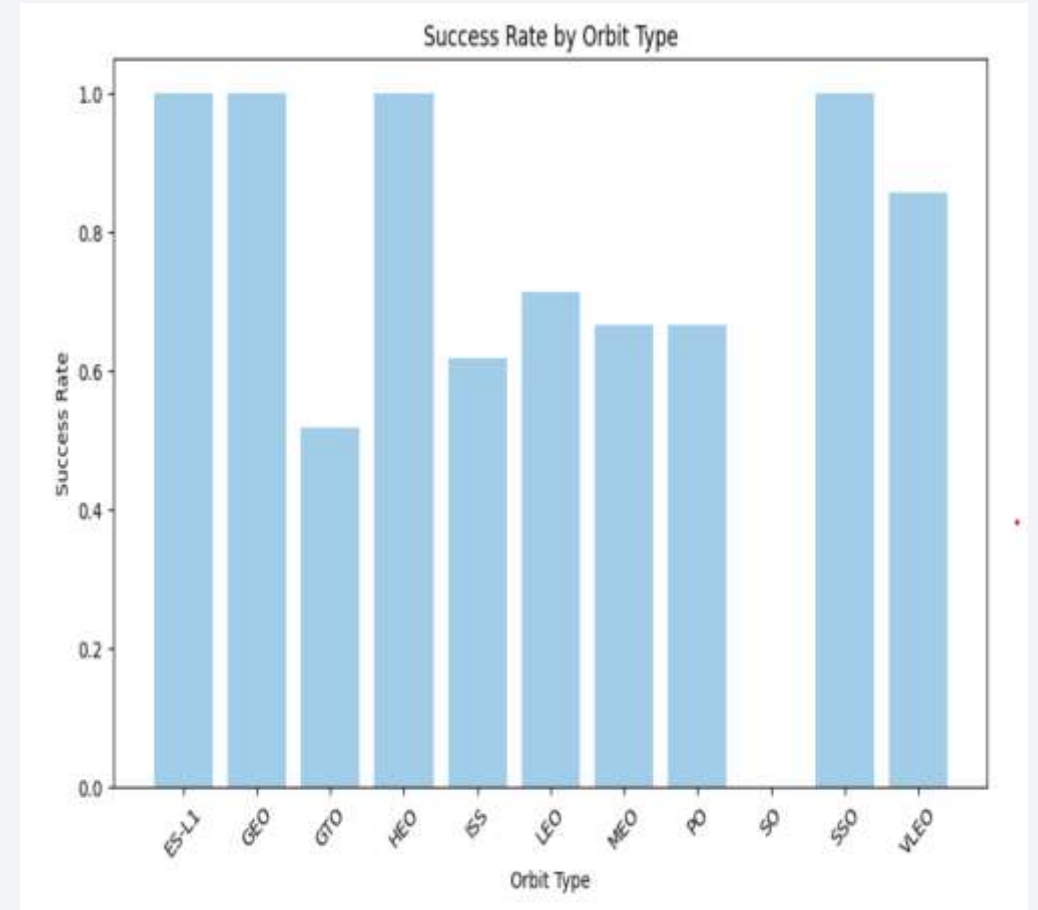
Displays how different payload weights are distributed across different launch sites. Helps to identify if certain sites are used for heavier payloads.



Success Rate vs. Orbit Type

Explanation:

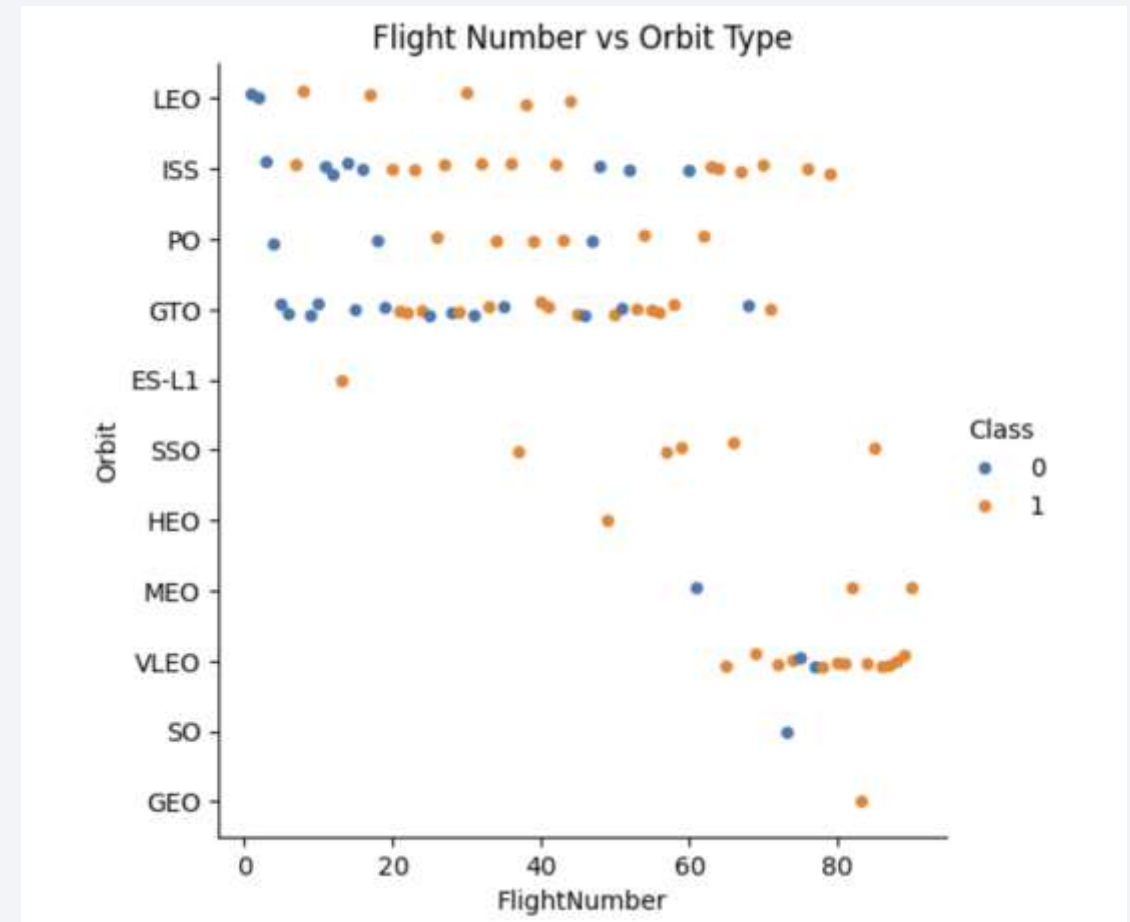
Compares the success rates of launches across different orbit types. Helps determine which orbits are more reliable in terms of **landing success**.



Flight Number vs. Orbit Type

Explanation:

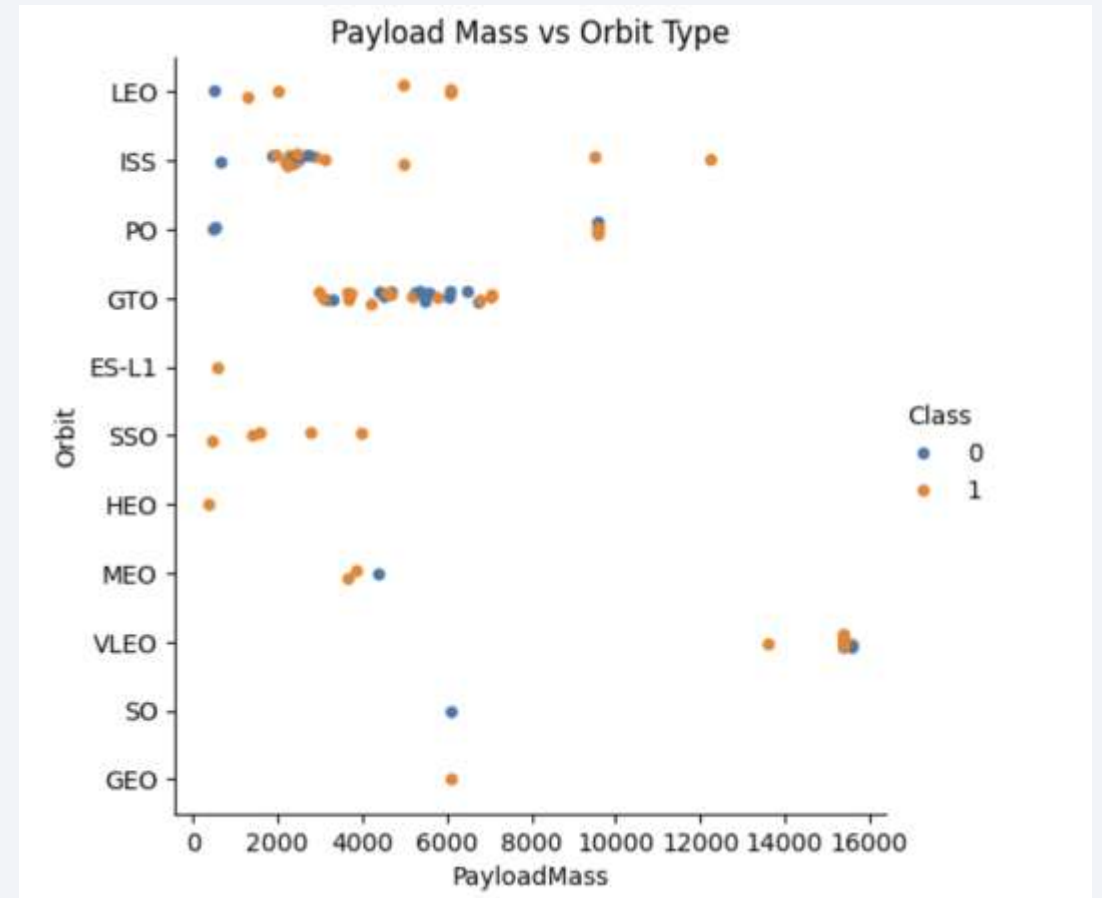
Visualizes trends between flight numbers and orbit types to analyze how frequently different orbits are targeted in SpaceX missions.



Payload vs. Orbit Type

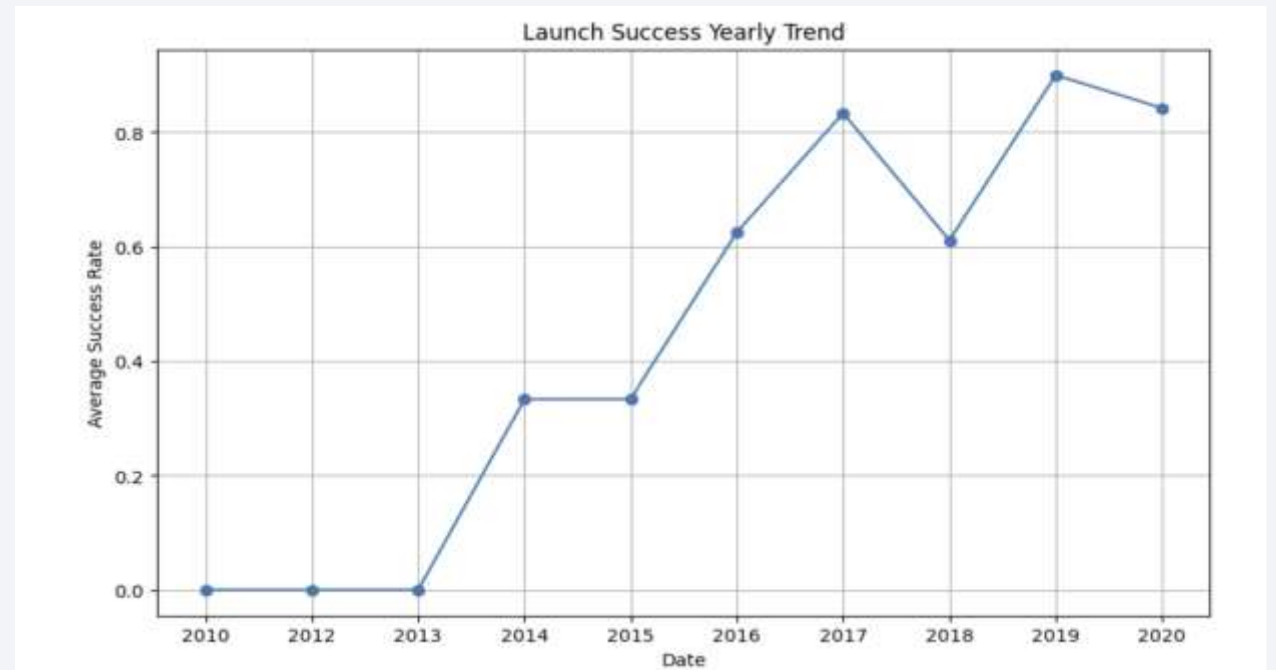
Explanation:

Shows how payload mass varies with different orbit types. Helps in understanding **payload** trends for specific mission types.



Launch Success Yearly Trend

- **Explanation:**
Illustrates the improvement in SpaceX's launch success rate over the years. Clear upward trend reflects operational efficiency over time.



All Launch Site Names

Explanation:

Extracted a list of **all unique launch site names** used by SpaceX for Falcon 9 missions.

Display the names of the unique launch sites in the space mission

In [19]: `%sql SELECT Distinct Launch_Site from SPACEXTABLE;`

* sqlite:///my_data1.db
Done.

Out[19]:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Explanation:

Filtered launch sites to **only include those starting with 'CCA'**

Display 5 records where launch sites begin with the string 'CCA'

```
In [21]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Out[21]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- **Explanation:**
Calculated the **total payload mass carried by NASA missions** using Falcon 9 boosters. Helps assess NASA's payload contributions.

```
Display the total payload mass carried by boosters launched by NASA (CRS)

In [25]: %sql select sum("PAYLOAD_MASS_KG_") as sum from SPACEXTABLE where Customer="NASA (CRS)"

* sqlite:///my_data1.db
Done.

Out[25]:  sum
         45596
```

Average Payload Mass by F9 v1.1

- **Explanation:**
Computed the **average payload mass for Falcon 9 v1.1 booster version** to analyze that model's payload capabilities.

Display average payload mass carried by booster version F9 v1.1

```
In [29]: %sql select avg("PAYLOAD_MASS_KG") as average from SPACEXTABLE where "Booster_Version"="F9 v1.1";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[29]: average
```

```
2928.4
```

First Successful Ground Landing Date

- **Explanation:**
Queried the **earliest date of a successful Falcon 9 first-stage ground landing** to identify milestones in SpaceX's landing success.

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
In [30]: %sql select min("date") as first_date from SPACEXTABLE where "Landing_Outcome"="Success (ground pad)"
```

```
* sqlite:///my_data1.db  
Done.
```

```
Out[30]: first_date
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- **Explanation:**
Filtered for **successful drone ship landings** where payload was **between 4000–6000 kg**, showing launches with challenging yet successful recoveries.

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [32]: %%sql select "Booster_Version" from spacetable WHERE "Landing_Outcome" = 'Success (drone ship)'
AND "PAYLOAD_MASS_KG_" > 4000
AND "PAYLOAD_MASS_KG_" < 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[32]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Total Number of Successful and Failure Mission Outcomes

- **Explanation:**
Calculated the **count of successful and failed Falcon 9 launches** overall to get a summary of launch performance.

List the total number of successful and failure mission outcomes

```
In [43]: %sql select "Mission_Outcome",count(*) as total from spacetable group by "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Out[43]:
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- **Explanation:**
Identified the **Falcon 9 booster version that carried the heaviest payload**, indicating SpaceX's high-capacity launch capability.

```
List all the booster_versions that have carried the maximum payload mass. Use a subquery.
```

```
In [48]: %%sql SELECT "Booster_Version"
FROM spacetable
WHERE "PAYLOAD_MASS_KG_" = (
    SELECT MAX("PAYLOAD_MASS_KG_")
    FROM spacetable
);
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[48]:
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- **Explanation:**

Filtered to show **failures during drone ship landings in 2015**, useful for identifying specific failures during SpaceX's operational improvements.

List the records which will display the month names, failure landing_outcomes in drone ship, booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
In [64]: %sql
SELECT substr("Date", 6, 2) AS Month, "Landing_Outcome", "Booster_Version", "Launch_Site"
FROM spaceXtable
WHERE "Landing_Outcome" = 'Failure (drone ship)'
AND substr("Date", 0, 5) = '2015';
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[64]:
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **Explanation:**
Ranked the **frequency of landing outcomes (success/failure types) from 2010 to 2017**, highlighting the most common landing results.

```
Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
```

```
In [65]: %%sql SELECT "Landing_Outcome", COUNT(*) AS Outcome_Count
FROM spacetable
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Outcome_Count DESC;

* sqlite:///my_data1.db
Done.
```

```
Out[65]:
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

<All Launch Sites - Global Map>

- **Explanation:**
Displays all SpaceX launch sites on a world map. Useful to visually confirm the global distribution of launch operations.



<Launch Outcomes with Colored Markers>

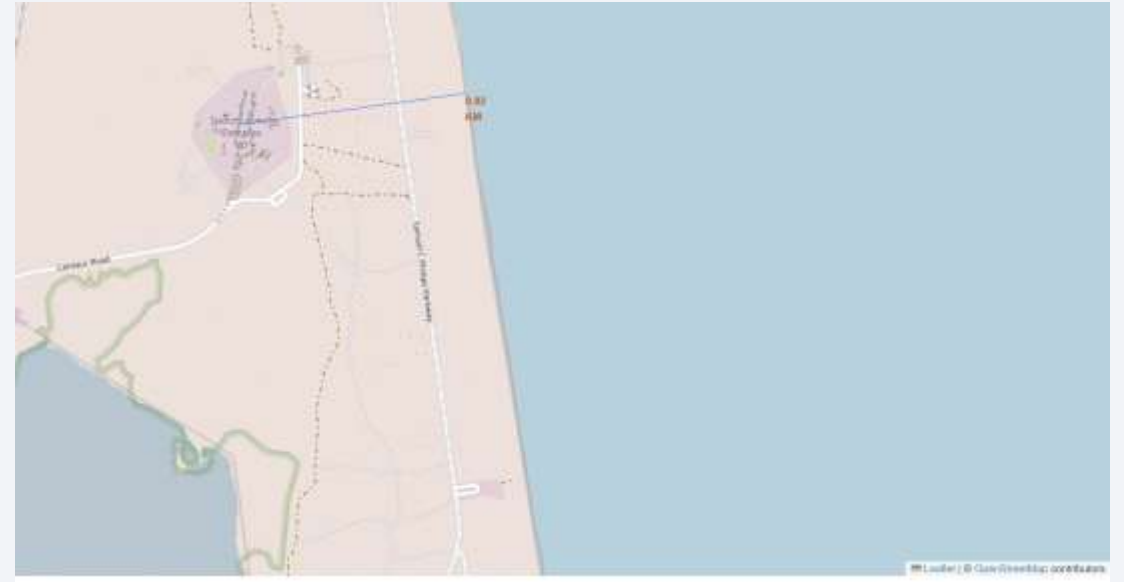
- **Explanation:**
Launch sites are marked with colored icons representing success (**green**) or failure (**red**). Helps identify which sites have higher success rates visually.

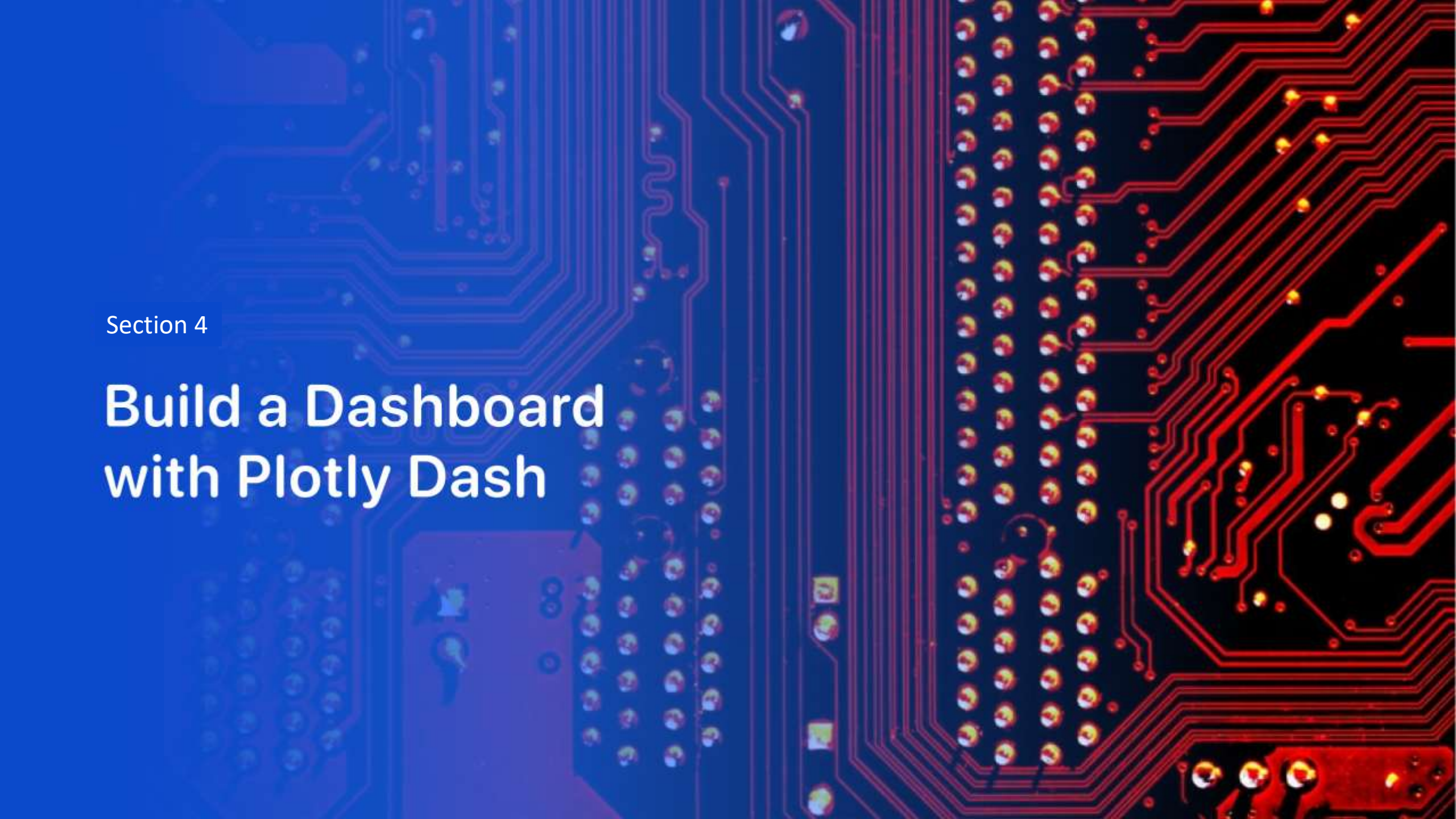


<Selected Launch Site with Proximity Lines>

- **Explanation:**

Focuses on a specific launch site with lines showing proximity to nearby infrastructure (railways, highways, [coastlines](#)). Useful for operational planning analysis.





Section 4

Build a Dashboard with Plotly Dash

<Launch Success Count for All Sites (Pie Chart)>

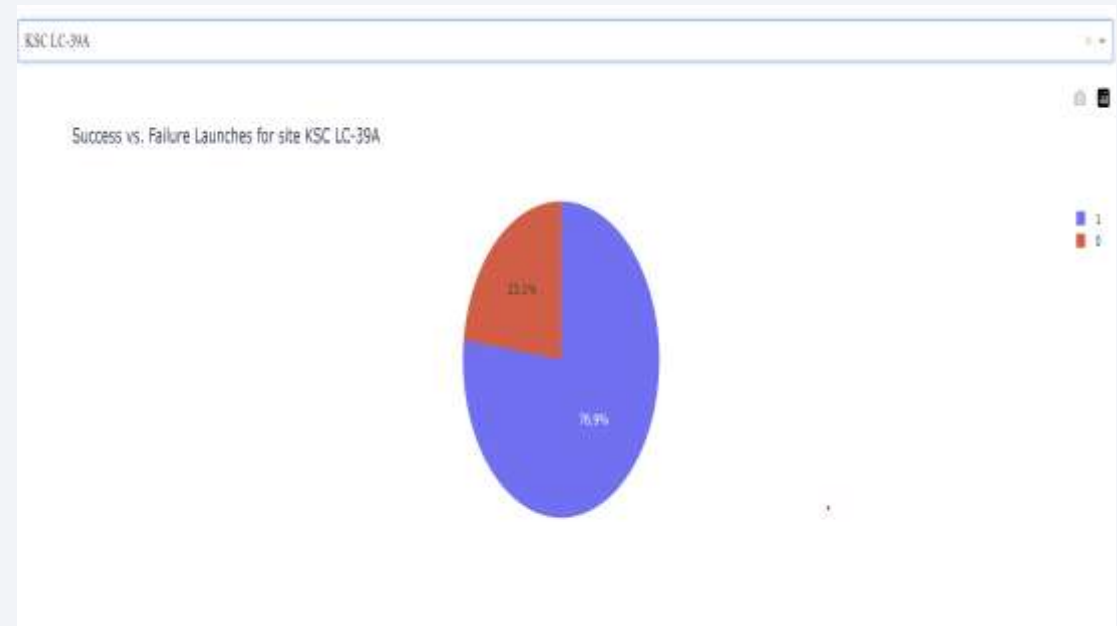
- **Explanation:**

Displays the **total number of successful launches across all launch sites**.
Helps **visualize** the **overall success ratio** for SpaceX Falcon 9 launches.



<Launch Success Count for Selected Site (Pie Chart)>

- **Explanation:**
Shows the **success vs. failure ratio** for a **selected KSC LC-39 A launch site**. Useful for comparing **individual site performance** against the total success rate.

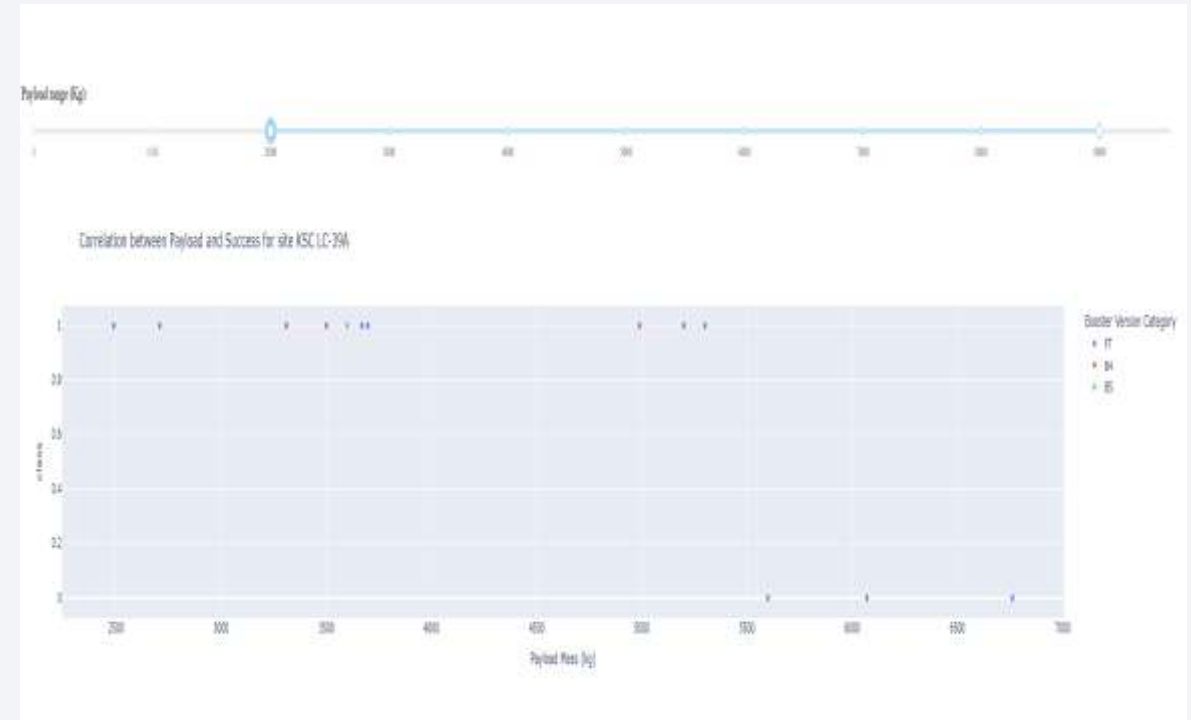


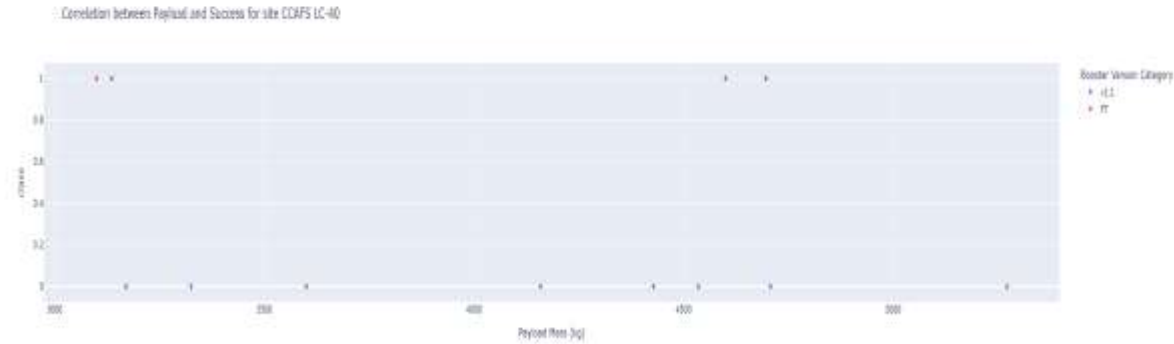
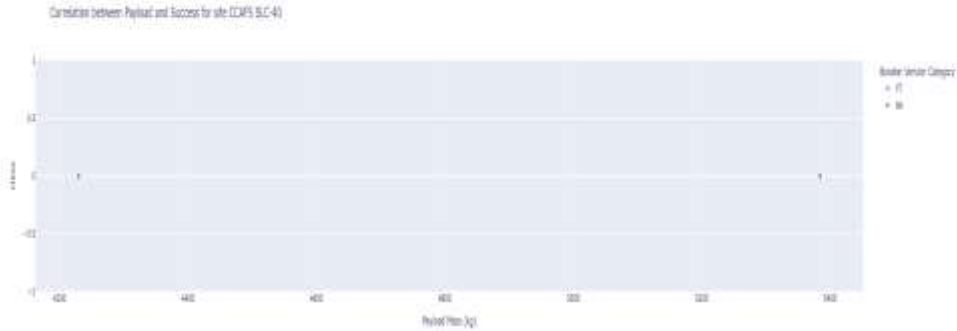
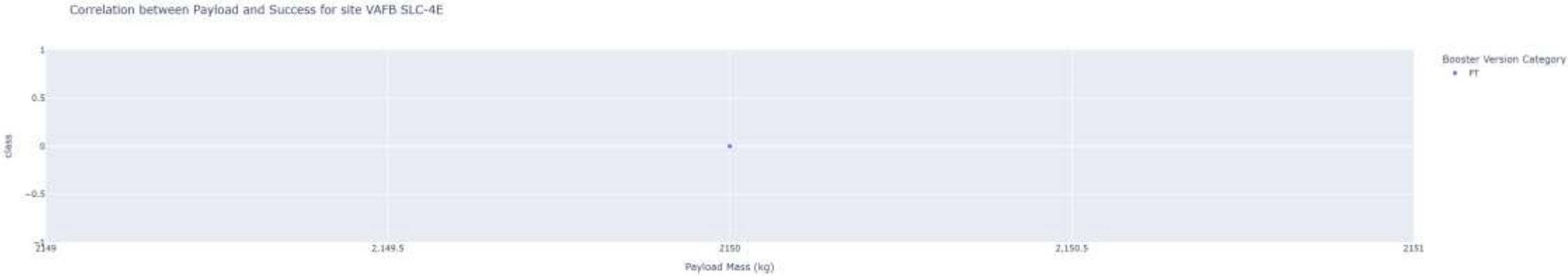
<Payload and success>

- **Explanation:**

Plots **payload mass vs. launch success** for all launch sites.

Observation: Payloads in the **4000–6000 kg range** had **higher success rates**, especially with certain booster versions like **F9 FT**.







Section 5

Predictive Analysis (Classification)

Classification Accuracy

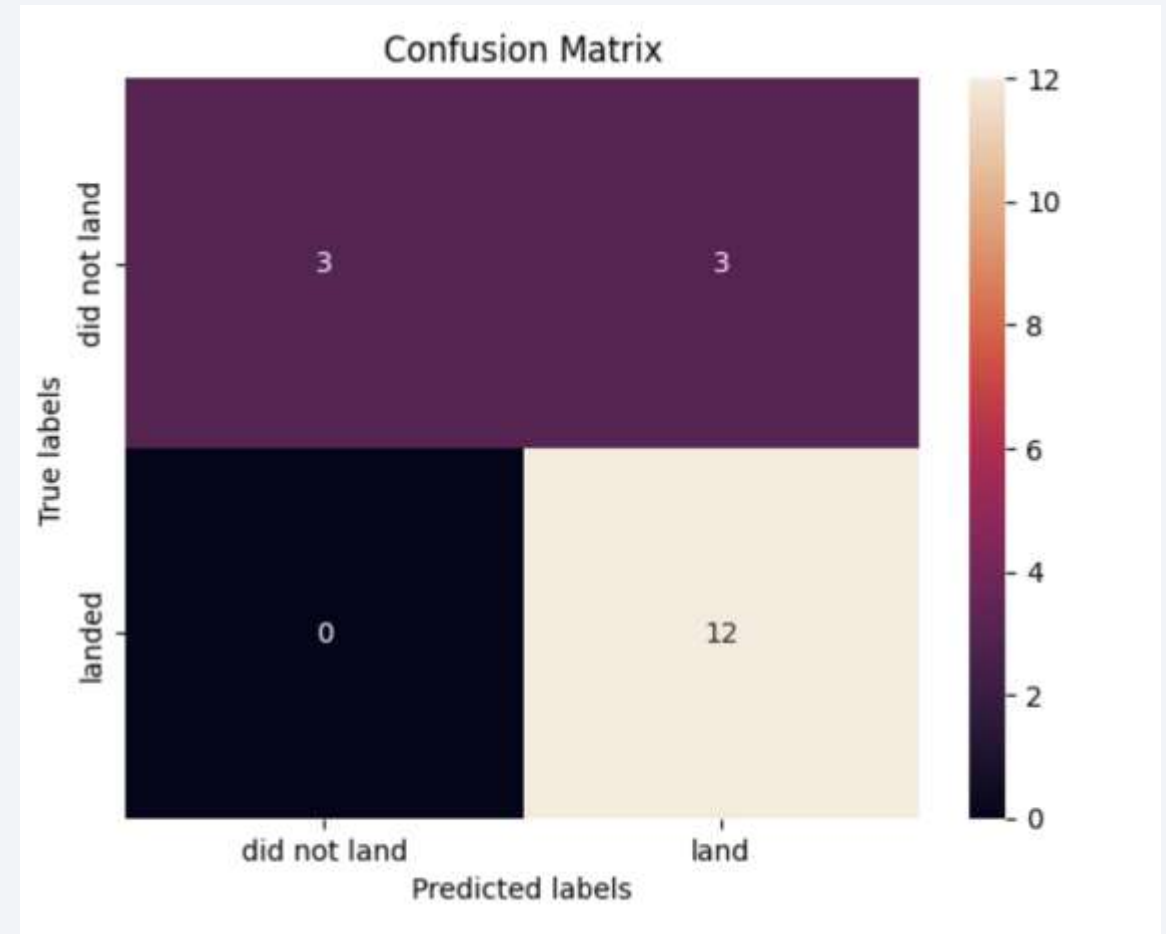
- **Some models performed similarly,**
with **slightly different accuracy scores**
but no single model showed clear
dominance.

Confusion Matrix

Explanation:

Confusion matrix of **one of the tested models** (SVM).

Visualizes **correct and incorrect predictions** for success and failure launches, helping evaluate model reliability.



Conclusions

- Successfully completed **end-to-end data analysis and prediction** on SpaceX Falcon 9 launch data.
- Collected data via **API, Web Scraping, and SQL**, and **cleaned** using data wrangling techniques.
- Applied **EDA, visualization, and interactive dashboards** to extract insights.
- **Built multiple machine learning models** → Found some **models performed similarly** in predicting landing success.
- Results provide valuable insights to **support SpaceX's operational planning and cost estimation**.

Appendix

- **Relevant Assets**
- **Python Code:** API collection, web scraping, EDA, ML models (Logistic Regression, SVM, Decision Tree)
- **SQL Queries:** Launch sites, payload analysis, success/failure counts
- **Charts:** Scatter plots, bar charts, pie charts, trend lines
- **Interactive Maps:** Folium → launch sites with success/failure markers
- **Dashboard:** Plotly Dash → Interactive filters and graphs
- **Datasets:** API data, scraped SpaceX data, SQLite database
- **Notebook Outputs:** Available in GitHub repo
-  **GitHub Repository:**
https://github.com/jatharva04/SpaceX_DataScience_Project

Thank you!

